

Annamalai Kalyanasundaram

Software Developer | GenAI Specialization (In Progress) | AWS Solutions Architect

Bengaluru, India · +91 78389 71194 · annamalai.k97@hotmail.com
linkedin.com/in/annamalai-kalyanasundaram

SUMMARY

Software engineer with ~2 years delivering production cloud and API systems, currently completing a Generative and Agentic AI specialization (expected October 2026). Building specialization capstones in LangGraph/LangChain RAG and agentic workflows, with planned packaging as a containerized FastAPI service. Production background includes REST APIs, React, MySQL, AWS, Docker, and CI/CD. AWS Certified Solutions Architect – Associate.

TECHNICAL SKILLS

Languages: Python, JavaScript / Node.js, SQL, Bash, Kotlin
Backend & Frontend: Node.js, Express.js, RESTful APIs, React.js, MySQL
Cloud & DevOps: AWS (EC2, VPC, S3, IAM, Lambda, ECS, RDS, CloudWatch), Docker, Jenkins, GitHub Actions, Git, Azure AD
Mobile: Kotlin, Jetpack Compose, Android SDK
GenAI (in progress†): LangChain, LangGraph, CrewAI, RAG, vector databases (FAISS, ChromaDB, Pinecone, Milvus), RAGAS, LangSmith, Hugging Face, PyTorch, Transformers, prompt engineering

† Generative & Agentic AI specialization and capstones — expected completion October 2026. FastAPI and Kubernetes are learning targets for the lead capstone, not production experience.

EXPERIENCE

Software Developer Jun 2024 – Jul 2026

Arche Global Pvt Ltd — Bengaluru, India

- Enhanced Pulse, a cloud operations platform for hybrid cloud environments, driving a ~30% improvement in operational efficiency through automated monitoring and streamlined workflows.
- Designed and maintained 15+ RESTful API endpoints (Node.js, Express.js) for asset management, ticketing, and monitoring under high-frequency read/write traffic.
- Established CI/CD workflows that reduced deployment time by streamlining release cycles and minimizing manual intervention across staging and production.
- Optimized MySQL database queries and schema design for high-frequency operations, improving data retrieval performance.
- Built responsive web interfaces in React.js for asset management, ticketing, and real-time cloud monitoring dashboards.
- Collaborated with cross-functional DevOps and infrastructure teams to deliver sprint commitments in an Agile environment.

GENAI SPECIALIZATION & CAPSTONES

Advanced Generative and Agentic AI In progress · Expected Oct 2026

Naresh Technologies

- Foundations:** Python for AI, neural networks, PyTorch, Transformer architecture, Hugging Face, multimodal LLMs, prompt engineering and guardrails.
- RAG systems:** Chunking, embeddings, ChromaDB, Pinecone, FAISS, Milvus, conversational RAG, RAGAS evaluation, Corrective RAG, Self-RAG, hybrid search, GraphRAG.
- Agentic AI:** LangChain, CrewAI, LangGraph (state graphs, persistence, human-in-the-loop), MCP, A2A, LangSmith observability.
- Optimization:** PEFT fine-tuning (LoRA/QLoRA), quantization, vLLM (coursework).

Agentic RAG Advisor — Lead Capstone In progress · Target Oct 2026

Planned stack: LangGraph · LangChain · ChromaDB / FAISS · RAGAS · LangSmith · FastAPI · Docker

- Designing a multi-agent advisor for a regulated domain: LangGraph orchestration with memory over a RAG pipeline (ingestion, chunking, embeddings, vector search, prompt assembly), with retrieval tuning, guardrails, RAGAS evaluation, and LangSmith tracing.
- Intended packaging as a containerized FastAPI service with REST endpoints, request/response validation, structured error handling, and basic observability.

Additional Capstones In progress · Target Oct 2026

- Domain-aware LLM chatbot with switching between open-source (Mistral) and API models.
- RAG-powered customer support assistant over internal knowledge bases.
- Agentic market-insight generation using web search tools.

ADDITIONAL PROJECTS

BIAL Plus — Internal Employee Mobile App
Kotlin · Jetpack Compose · Azure AD · Microsoft Intune · CI/CD

Production · Google Play

- Developed and shipped the official internal mobility platform for Bangalore International Airport employees, centralizing HCM, shift management, and HR services.
- Built the app in Kotlin/Jetpack Compose; integrated Azure AD single sign-on and Microsoft Intune (MDM/MAM); resolved critical auth blockers to support 24/7 airport operations.

Pulse — Cloud Operations Platform
React.js · Node.js · Express.js · MySQL · AWS · RESTful APIs

Arche Global

- Architected a cloud operations platform for hybrid environment management — infrastructure visibility and incident response.
- Designed 15+ RESTful API endpoints and a MySQL-backed data layer with optimized queries; built real-time monitoring dashboards in React.js.

ArcheOne — Productivity Platform (Android)
Kotlin · Android SDK · RESTful APIs · Push Notifications

Arche Global

- Delivered end-to-end features (Business Card, SOS, Desccart, TravelDesk, Helpdesk) with real-time push notifications and optimized REST integrations across devices.

EDUCATION

Bachelor of Technology (B.Tech), Engineering
Sri Venkateswara College of Engineering, Chennai

2021

Advanced Generative and Agentic AI
Naresh Technologies

In progress · Expected Oct 2026

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** — valid through March 2027
- **APTECH Certified Python Developer**

LANGUAGES

English (Fluent) · Hindi (Fluent) · Tamil (Native)